

UNIT-IV

VECTOR DIFFERENTIATION

Consider a 3-dimensional region (R), -for each point P(x,y,z) if we associate a scalar (ϕ) - [is a function $\phi(x,y,z)$] such a function ϕ is called scalar point function & the region R is called scalar field

Ex: Temperature distribution in a medium

If we associate a vector $\vec{f}$ ($\vec{f}(x,y,z)$) - at each point then $\vec{f}$ is called vector point function

Ex: 1 The velocity of a particle in a moving fluid
2 Gravitational field

Vector differential operator (∇): Del &

$\rightarrow \nabla$ is denoted by Del. Defined as $\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$

$\rightarrow$ We operate ∇ on scalar point functions & vector point functions

Gradient: - The gradient of a scalar ϕ is defined by $\nabla \phi$ (a)

grad ϕ is defined as $\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$

Physical Significance of Gradient

(1) $\nabla \phi$ is normal to the surface ($\phi = c$)

(2) $\nabla \phi$ Gives max rate of change in ϕ

(3) $|\nabla \phi|$ is the max change

$$\therefore \sqrt{\left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2 + \left(\frac{\partial \phi}{\partial z}\right)^2}$$



Directional derivative

(1) The directional derivative of a scalar $\phi = c$ is denoted by $\nabla \phi \cdot \hat{e}$

where $\hat{e}$ = unit vector in given direction

$\nabla \phi$ = gradient of ϕ

(19) Find the gradient of surface $\phi = \log(x^4 + y^4 + z^4)$

Sol $\nabla\phi = \text{grad } \phi$

$$= \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$$

$$= \frac{2x}{x^4 + y^4 + z^4} \hat{i} + \frac{2y}{x^4 + y^4 + z^4} \hat{j} + \frac{2z}{x^4 + y^4 + z^4} \hat{k}$$

$$= \frac{2}{x^4 + y^4 + z^4} [x\hat{i} + y\hat{j} + z\hat{k}]$$

$$= \frac{2\vec{r}}{|\vec{r}|^4}$$

(20) Find the unit Normal to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$

Sol $\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$

$$= (3x^2 + 3yz)\hat{i} + (3y^2 + 3xz)\hat{j} + 3xy\hat{k} \text{ which is normal to } \phi$$

unit normal to the surface

$$= \frac{\nabla\phi}{|\nabla\phi|}$$

$$\text{Here } \nabla\phi_{(1,2,-1)} = -3\hat{i} + 9\hat{j} + 6\hat{k}$$

$$\therefore \hat{n} = \frac{[\nabla\phi]_p}{|\nabla\phi|} = \frac{-3\hat{i} + 9\hat{j} + 6\hat{k}}{\sqrt{9 + 81 + 36}} = \frac{-3\hat{i} + 9\hat{j} + 6\hat{k}}{\sqrt{126}}$$

Q6) Find the angle b/w the two surfaces $x+y+z=9$ &

$x^2+y^2-z=3$ at the point $(2, -1, 2)$

Sol) Given that

$$\phi = x^2 + y^2 + z - 9$$

$$\psi = x^2 + y^2 - z - 3$$

Let θ be the angle b/w ϕ & ψ

$$\text{then } \cos \theta = \frac{\nabla \phi \cdot \nabla \psi}{|\nabla \phi| |\nabla \psi|}$$

$$\nabla \phi = 2x\vec{i} + 2y\vec{j} + \vec{k}$$

$$\nabla \phi|_{(2, -1, 2)} = 4\vec{i} - 2\vec{j} + \vec{k}$$

$$\nabla \psi = 2x\vec{i} + 2y\vec{j} - \vec{k}$$

$$\nabla \psi|_{(2, -1, 2)} = 4\vec{i} - 2\vec{j} - \vec{k}$$

$$\cos \theta = \frac{16 + 4 - 4}{\sqrt{16+4+1} \sqrt{16+4+1}} = \frac{16}{8\sqrt{21}}$$

$$\theta = \cos^{-1} \left(\frac{2}{\sqrt{21}} \right)$$

Q2 Find the directional derivative of $\phi = x^2y^2 + xyz^2$ at the point $(1, -2, 1)$ in the direction of $2\bar{i} - \bar{j} - 2\bar{k}$

Sol Given $\phi = x^2y^2 + xyz^2$

$$\text{now } \nabla\phi = (2xy^2 + yz^2)\bar{i} + (x^2z + 0)\bar{j} + (x^2y + 2xz)\bar{k}$$

$$(\nabla\phi)_{(1, -2, 1)} = (-4 + 4)\bar{i} + \bar{j} + 6\bar{k}$$

$$\nabla\phi|_{(1, -2, 1)} = 0\bar{i} + \bar{j} + 6\bar{k}$$

$$\therefore \text{Directional derivative} = \nabla\phi|_P \cdot \hat{e}$$

$$= (\bar{j} + 6\bar{k}) \cdot \left[\frac{2\bar{i} - \bar{j} - 2\bar{k}}{\sqrt{4+1+4}} \right]$$

$$= (\bar{j} + 6\bar{k}) \cdot \left[\frac{2\bar{i} - \bar{j} - 2\bar{k}}{3} \right]$$

$$= \frac{-1 - 12}{3} = \frac{-13}{3}$$

Q6 The temperature of points in space $T(x, y, z) = x^2y^2 - z^2$, a mosquito is located at the point $(1, 1, 2)$ desires to fly in such a direction that it will get warm as soon as possible, in what direction should we move?

Sol Given that $T = x^2y^2 - z^2$

$$\text{now } \nabla T = 2x\bar{i} + 2y\bar{j} - 2\bar{k}$$

Which is normal to T

The mosquito should move in the direction

$$= \nabla T|_P / |\nabla T|$$

$$= \frac{2\bar{i} + 2\bar{j} - 2\bar{k}}{3}$$

(P6) Find the directional derivative of $\phi = xy^2 + yz^3$ at the point $P(2, -1, 1)$ in the direction of normal to the surface $x \log z - y^2 = -4$ at the point $Q(-1, 2, 1)$

Sol Given that

$$\phi = xy^2 + yz^3$$

$$\nabla \phi = y^2 \bar{i} + (2xy + z^3) \bar{j} + (3yz^2) \bar{k}$$

$$\nabla \phi|_{P(2, -1, 1)} = \bar{i} - 3\bar{j} - 3\bar{k}$$

Take, $\psi = x \log y - y^2 + y = \bar{e}$

$$\nabla \psi = \log z \bar{i} - 2y \bar{j} + \frac{1}{z} \bar{k}$$

$$\nabla \psi|_{Q(-1, 2, 1)} = 0\bar{i} - 4\bar{j} - \bar{k}$$

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \cos^2 \theta &= 1 - 3 \\ \cos^2 \theta &= -2 \end{aligned}$$

Let $\hat{e} = \frac{\Delta \psi}{|\Delta \psi|} = \frac{-4\bar{j} - \bar{k}}{\sqrt{16+1}} = \frac{-4\bar{j} - \bar{k}}{\sqrt{17}}$

$\therefore$ required directional derivative

$$= \nabla \phi|_P \cdot \hat{e}$$

$$= (\bar{i} - 3\bar{j} - 3\bar{k}) \cdot \left(\frac{-4\bar{j} - \bar{k}}{\sqrt{17}} \right)$$

$$= \frac{12+3}{\sqrt{17}} = \frac{15}{\sqrt{17}}$$

Del Applied to vector

(Pb) P.T $\nabla(r^n) = nr^{n-2}(\vec{r})$

Where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ & $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$

Here

$$r = \sqrt{x^2 + y^2 + z^2} = |\vec{r}|$$

$$\frac{dr}{dx} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \cdot 2x = \frac{x}{r}$$

$$\text{or } \frac{\partial r}{\partial y} = \frac{y}{r} \quad ; \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\text{L.H.S} = \nabla(r^n)$$

$$= \frac{\partial}{\partial x}(r^n)\vec{i} + \frac{\partial}{\partial y}(r^n)\vec{j} + \frac{\partial}{\partial z}(r^n)\vec{k}$$

$$= nr^{n-1} \left[\frac{x}{r}\vec{i} + \frac{y}{r}\vec{j} + \frac{z}{r}\vec{k} \right]$$

$$= nr^{n-2}(\vec{r}) = \text{R.H.S}$$

(Pb) P.T $\nabla\left(\frac{1}{r}\right) = \frac{-\vec{r}}{r^3}$

sol $\nabla\left(\frac{1}{r}\right) = \frac{\partial}{\partial x}\left(\frac{1}{r}\right)\vec{i} + \frac{\partial}{\partial y}\left(\frac{1}{r}\right)\vec{j} + \frac{\partial}{\partial z}\left(\frac{1}{r}\right)\vec{k}$

$$= \frac{-1}{r^2} \left[\frac{x\vec{i} + y\vec{j} + z\vec{k}}{r} \right]$$

$$= \frac{-\vec{r}}{r^3} = \text{R.H.S}$$

Del Applied to vector point functions

Divergence: The Divergence of a vector point function $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ is denoted by $\text{div } \vec{F}$ (or) $\nabla \cdot \vec{F}$ & defined as

$$\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \cdot (F_1\vec{i} + F_2\vec{j} + F_3\vec{k})$$

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \quad (\text{scalar})$$

Physical Significance of Divergence:

$\text{curl } \vec{F} = 0$
 $\nabla \times \vec{F} = 0$

① Divergence of a vector gives the rate per unit volume at which the physical quantity issuing from that point

Ex: If $\vec{v}$ represents velocity of fluids, then $\text{div } \vec{v}$ or $\nabla \cdot \vec{v}$ gives the weight at which the fluid is coming at a point in unit volume

② If divergence of $\vec{F}$ is zero then $\vec{F}$ is called solenoidal vector field $\text{div } \vec{v}$

curl: The curl of the vector field $\vec{F}$ is denoted by $\text{curl } \vec{F}$

or $\nabla \times \vec{F}$ & defined as $\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$

$$\nabla \times \vec{F} = \vec{i} \left[\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right] + \vec{j} \left[\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right] + \vec{k} \left[\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right]$$

Physical significance of curl:

① If $\vec{\omega}$ be the angular velocity of a rigid body rotating about a fixed axis & $\vec{v}$ be the velocity any particle then

$$\vec{\omega} = \frac{1}{2} \text{curl } \vec{v}$$

* ② If curl of $\vec{F}$ is equal to zero, then $\vec{F}$ is called irrotational vector field.

$$\begin{cases} \nabla \cdot (\nabla \phi) = \nabla^2 \phi \\ \nabla \cdot (\nabla \times \vec{F}) = 0 \end{cases} \quad \begin{cases} \nabla \times (\nabla \phi) = \vec{0} \\ \nabla \times (\nabla \times \vec{F}) = \nabla (\nabla \cdot \vec{F}) - \nabla^2 \vec{F} \end{cases}$$

(16) If $\vec{F} = xy^2\vec{i} + 2xyz\vec{j} + 3yz^2\vec{k}$ then find $\text{div } \vec{F}$ & $\text{curl } \vec{F}$ at the point $(1, -1, 1)$

Sol Given that,

$$\begin{aligned}\nabla \cdot \vec{F} = \text{div } \vec{F} &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\ &= \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(2xyz) + \frac{\partial}{\partial z}(-3yz^2) \\ &= y^2 + 2xz - 6yz\end{aligned}$$

$$\text{Now } (\nabla \cdot \vec{F})_{(1, -1, 1)} = 1 + 2 + 6 = 9$$

$$\nabla \times \vec{F} = \text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & 2yz & -3yz^2 \end{vmatrix}$$

$$\begin{aligned}&= \vec{i} \left[-\frac{\partial}{\partial y}(3yz^2) - \frac{\partial}{\partial z}(2xy) \right] + \vec{j} \left[-\frac{\partial}{\partial z}(xy^2) + \frac{\partial}{\partial x}(-3yz^2) \right] + \vec{k}(xy^2 - 2xy) \\ &= \vec{i}(-3z^2 - 2xy) + \vec{k}(xy^2 - 2xy)\end{aligned}$$

$$\text{Now } (\text{curl } \vec{F})_{(1, -1, 1)} = -\vec{i} - 2\vec{k}$$

(17) Find $\text{div } \vec{F}$ & $\text{curl } \vec{F}$ where $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$

Sol Given that

$$\vec{F} = \underbrace{(3x^2 - 3yz)}_{F_1}\vec{i} + \underbrace{(3y^2 - 3xz)}_{F_2}\vec{j} + \underbrace{(3z^2 - 3xy)}_{F_3}\vec{k}$$

$$\begin{aligned}\nabla \cdot \vec{F} = \text{div } \vec{F} &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\ &= (6x - 0) + 6y + 6z \\ &= 6x + 6y + 6z = 6(x + y + z)\end{aligned}$$

$$\begin{aligned}\nabla \times \vec{F} = \text{curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 3yz & 3y^2 - 3xz & 3z^2 - 3xy \end{vmatrix} \\ &= \vec{i}(-3x + 3x) + \vec{j}(3y + 3y) + \vec{k}(-3z + 3z) \\ &= 0\vec{i} + 0\vec{j} + 0\vec{k} = \vec{0}\end{aligned}$$

Q6) If $\vec{F} = (x+3y)\vec{i} + (y-2z)\vec{j} + (x+pz)\vec{k}$ is solenoidal then find P. Hence find P.

Sol Suppose $\vec{F}$ is solenoidal

$$\text{i.e., } \text{div } \vec{F} = 0$$

$$\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0$$

$$1 + 1 + p = 0$$

$$\boxed{p = -2}$$

Q7) If $\vec{F} = (x+y+1)\vec{i} + \vec{j} - (x+y)\vec{k}$ then show that $\vec{F} \cdot \text{curl } \vec{F} = 0$

Sol $\nabla \times \vec{F} = \text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x+y+1) & 1 & -(x+y) \end{vmatrix}$

$$= \vec{i}(-1) - \vec{j}(-1-0) + \vec{k}(0-1)$$

$$= -\vec{i} + \vec{j} - \vec{k}$$

$$\text{L.H.S} = \vec{F} \cdot \text{curl } \vec{F} = (x+y+1)(-1) + (1)(1) + (x+y)$$

$$= -x-y-1+1+x+y$$

$$= 0 = \text{R.H.S}$$

*

Q8) Prove that $\nabla \cdot (\nabla^n \vec{r}) = (n+3)\nabla^n$ & hence so that

$\vec{r}/r^3$ is solenoidal

Sol L.H.S = $\nabla \cdot [\nabla^n (x\vec{i} + y\vec{j} + z\vec{k})]$

$$= \frac{\partial}{\partial x}(\nabla^n x) + \frac{\partial}{\partial y}(\nabla^n y) + \frac{\partial}{\partial z}(\nabla^n z)$$

$$= \nabla^n \cdot 1 + n \nabla^{n-1} \frac{\partial}{\partial x} + \nabla^n \cdot 1 + n \nabla^{n-1} \frac{\partial}{\partial y} + \nabla^n \cdot 1 + n \nabla^{n-1} \frac{\partial}{\partial z}$$

$$= 3\nabla^n + n \nabla^{n-1} \left[x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z} \right]$$

$$= 3\nabla^n + n \nabla^{n-1} \left[\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} \right]$$

$$= 3\nabla^n + n \nabla^n (x^2+y^2+z^2) = (n+3)\nabla^n = \text{R.H.S}$$

$$\therefore \nabla \cdot (r^n) = (n+3)r^n$$

$$\text{Put } (n = -3)$$

$$\text{i.e., } \nabla \cdot \left(\frac{r}{r^3} \right) = 0$$

$$\therefore \frac{r}{r^3} \text{ is solenoidal}$$

Del Applied twice to point functions

$$\textcircled{1} \nabla \cdot (\nabla f) = \nabla \cdot \left(\frac{\partial f}{\partial x} \bar{i} + \frac{\partial f}{\partial y} \bar{j} + \frac{\partial f}{\partial z} \bar{k} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial z} \right)$$

$$= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) f$$

$$= \nabla^2 f \quad (\text{scalar})$$

$$\textcircled{2} \nabla \times \nabla f = \bar{0} \quad [\text{curl}(\text{grad } f) = \bar{0}]$$

$$\text{So L.H.S} = \nabla \times \nabla f = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}$$

$$= \bar{i} \left[\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial y \partial z} \right] + \bar{j} \left[\frac{\partial^2 f}{\partial x \partial z} - \frac{\partial^2 f}{\partial x \partial z} \right] + \bar{k} \left[\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial x \partial y} \right]$$

$$= \bar{0} = \text{R.H.S}$$

$$\textcircled{3} \quad \text{Div}(\text{curl } \vec{F}) = 0$$

(or)

$$\nabla \cdot (\nabla \times \vec{F}) = 0$$

$$\text{Now } \nabla \cdot (\nabla \times \vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial F_1}{\partial x} & \frac{\partial F_2}{\partial y} & \frac{\partial F_3}{\partial z} \end{vmatrix}$$

$$= \vec{i} \left[\frac{\partial^2 F_3}{\partial y \partial z} - \frac{\partial^2 F_2}{\partial y \partial z} \right]$$

$$\text{L.H.S} = \nabla \cdot \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \vec{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \vec{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \vec{k} \right]$$

$$= \frac{\partial^2 F_3}{\partial y \partial z} - \frac{\partial^2 F_2}{\partial z \partial x} + \frac{\partial^2 F_1}{\partial z \partial y} - \frac{\partial^2 F_3}{\partial x \partial y} + \frac{\partial^2 F_2}{\partial x \partial z} - \frac{\partial^2 F_1}{\partial y \partial z}$$

$$= 0 = \text{R.H.S}$$

$$\textcircled{4} \quad \nabla \times (\nabla \times \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$$

$$\text{L.H.S} = \nabla \times (\nabla \times \vec{F})$$

$$= \nabla \times \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \vec{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \vec{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \vec{k} \right]$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) & \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) & \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \end{vmatrix}$$

$$= \vec{i} \left[\left(\frac{\partial^2 F_2}{\partial x \partial y} - \frac{\partial^2 F_1}{\partial y^2} \right) - \frac{\partial^2 F_1}{\partial z^2} + \frac{\partial^2 F_3}{\partial z \partial x} \right] + \vec{j} \left[\frac{\partial^2 F_2}{\partial x^2} - \frac{\partial^2 F_1}{\partial x \partial y} - \frac{\partial^2 F_3}{\partial y \partial z} + \frac{\partial^2 F_2}{\partial y^2} \right]$$

$$+ \vec{k} \left[\frac{\partial^2 F_1}{\partial z \partial x} - \frac{\partial^2 F_3}{\partial x^2} - \frac{\partial^2 F_3}{\partial y^2} + \frac{\partial^2 F_2}{\partial y \partial z} \right]$$

$$= \bar{i} \left[\frac{\partial f_1}{\partial x^1} + \frac{\partial f_2}{\partial x^2} + \frac{\partial f_3}{\partial x^3} - \left(\frac{\partial f_1}{\partial x^1} + \frac{\partial f_2}{\partial x^2} + \frac{\partial f_3}{\partial x^3} \right) \right] \\ + \bar{j} \left[\frac{\partial f_2}{\partial x^1} + \frac{\partial f_3}{\partial x^2} + \frac{\partial f_1}{\partial x^3} - \left(\frac{\partial f_2}{\partial x^1} + \frac{\partial f_3}{\partial x^2} + \frac{\partial f_1}{\partial x^3} \right) \right] \\ + \bar{k} \left[\frac{\partial f_3}{\partial x^1} + \frac{\partial f_1}{\partial x^2} + \frac{\partial f_2}{\partial x^3} - \left(\frac{\partial f_3}{\partial x^1} + \frac{\partial f_1}{\partial x^2} + \frac{\partial f_2}{\partial x^3} \right) \right]$$

$$= \bar{i} \frac{\partial}{\partial x} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) - \bar{i} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) f_1 \\ + \bar{j} \frac{\partial}{\partial y} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) - \bar{j} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) f_2 \\ + \bar{k} \frac{\partial}{\partial z} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) - \bar{k} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) f_3 \quad (17)$$

$$= \nabla (\nabla \cdot \vec{F}) - \nabla^2 \vec{F} = \text{R.H.S}$$

(18) $\nabla^2 f(x) = f''(x) + \frac{2}{x} f'(x)$

L.H.S = $\text{div} [\text{grad}(f(x))]$

$$= \nabla [f(x)]$$

$$= \left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right] (f(x))$$

$$x = \sqrt{x^2 + y^2 + z^2}$$

$$\frac{\partial x}{\partial x} = \frac{x}{x}$$

$$\frac{\partial x}{\partial y} = \frac{y}{x}$$

$$\frac{\partial x}{\partial z} = \frac{z}{x}$$

$$= \frac{\partial}{\partial x} \left[f'(x) \frac{\partial x}{\partial x} \right] + \frac{\partial}{\partial y} \left[f'(x) \frac{\partial x}{\partial y} \right] + \frac{\partial}{\partial z} \left[f'(x) \frac{\partial x}{\partial z} \right]$$

$$= \frac{\partial}{\partial x} \left[f'(x) \frac{x}{x} \right] + \frac{\partial}{\partial y} \left[f'(x) \frac{y}{x} \right] + \frac{\partial}{\partial z} \left[f'(x) \frac{z}{x} \right]$$

$$= f'(x) \cdot \left[\frac{x \cdot 1 - x \cdot \frac{\partial x}{\partial x}}{x^2} \right] + \frac{x}{x} f''(x) - \frac{\partial x}{\partial x}$$

$$+ f'(x) \cdot \left[\frac{x \cdot 1 - y \cdot \frac{\partial x}{\partial y}}{x^2} \right] + \frac{y}{x} f''(x) \frac{\partial x}{\partial y}$$

$$+ f'(x) \cdot \left[\frac{x \cdot 1 - z \cdot \frac{\partial x}{\partial z}}{x^2} \right] + \frac{z}{x} f''(x) \frac{\partial x}{\partial z}$$

$$= f'(x) \left[\frac{x^N - x^N}{x^3} \right] + f''(x) \frac{x^N}{x^N}$$

$$+ f'(x) \left[\frac{x^N - y^N}{x^3} \right] + f''(x) \frac{y^N}{x^N}$$

$$+ f'(x) \left[\frac{x^N - z^N}{x^3} \right] + f''(x) \frac{z^N}{x^N}$$

$$= f'(x) \left[\frac{x^N - x^N + x^N - y^N + x^N - z^N}{x^3} \right] + f''(x) \left[\frac{x^N + y^N + z^N}{x^N} \right]$$

$$= f'(x) + \frac{2}{x} f'(x) = R.H.S$$

(Pb) If $f = (x^2 + y^2 + z^2)^{-n}$ then find $\nabla \cdot \text{grad } f$ & determine 'n' if $\nabla \cdot \text{grad } f = 0$

Sol. Q.T. $f = (x^2 + y^2 + z^2)^{-n}$

~~#~~

$$f' = (-2n)x^{-2n-1}$$

$$f'' = (-2n)(-2n-1)x^{-2n-2}$$

N.K.T

$$\nabla^2 f(x) = f''(x) + \frac{2}{x} f'(x)$$

$$\Rightarrow \nabla^2 f = \text{div}(\text{grad } f)$$

$$= -2n(-2n-1)x^{-2n-2} + \frac{2}{x}(-2n)x^{-2n-1}$$

$$= (2n)(2n+1)x^{-2n-2} - 4nx^{-2n-2}$$

$$= x^{-2n-2} [(2n)(2n+1-2)]$$

$$= x^{-2n-2} [2n(2n-1)]$$

Suppose $\text{div grad } f = 0$

$$\Rightarrow \nabla^2 f = 0 \quad [2n(2n-1)] = 0$$

$$\Rightarrow \begin{cases} 2n=0 \\ n=0 \end{cases} \quad \begin{cases} 2n-1=0 \\ n=\frac{1}{2} \end{cases}$$

* (P6) Show that $\nabla \times (\nabla \times (\nabla \times (\nabla \times \vec{F}))) = \nabla^4 \vec{F}$ if $\vec{F}$ is solenoidal
Suppose $\vec{F}$ is solenoidal ; i.e., $\nabla \cdot \vec{F} = 0$

L.H.S

$$= \nabla \times \nabla \times (\nabla \times (\nabla \times \vec{F}))$$

$$= \nabla \times \nabla \times (\nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}) \quad \because [\nabla \cdot \vec{F} = 0]$$

$$= \nabla \times \nabla \times (-\nabla^2 \vec{F})$$

$$= -\nabla^2 [\nabla \times (\nabla \times \vec{F})]$$

$$= -\nabla^2 [\nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}]$$

$$= -\nabla^2 [-\nabla^2 \vec{F}]$$

$$= \nabla^4 \vec{F} = \text{R.H.S}$$

(P7) If f, g are 2 scalar functions then p.t

$$\text{div}(f \nabla g) = f \nabla^2 g + \nabla f \cdot \nabla g$$

or L.H.S = $\text{div}(f \nabla g)$

$$= \text{div} \left[f \frac{\partial g}{\partial x} \hat{i} + f \frac{\partial g}{\partial y} \hat{j} + f \frac{\partial g}{\partial z} \hat{k} \right]$$

$$= \nabla \cdot \left[f \frac{\partial g}{\partial x} \hat{i} + f \frac{\partial g}{\partial y} \hat{j} + f \frac{\partial g}{\partial z} \hat{k} \right]$$

$$= \frac{\partial}{\partial x} \left(f \frac{\partial g}{\partial x} \right) + \frac{\partial}{\partial y} \left(f \frac{\partial g}{\partial y} \right) + \frac{\partial}{\partial z} \left(f \frac{\partial g}{\partial z} \right)$$

$$= f \frac{\partial^2 g}{\partial x^2} + \frac{\partial f}{\partial x} \frac{\partial g}{\partial x} + f \frac{\partial^2 g}{\partial y^2} + \frac{\partial f}{\partial y} \frac{\partial g}{\partial y} + f \frac{\partial^2 g}{\partial z^2} + \frac{\partial f}{\partial z} \frac{\partial g}{\partial z}$$

$$= f \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) g + \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right) \cdot \left(\frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} + \frac{\partial g}{\partial z} \hat{k} \right)$$

$$= f \nabla^2 g + \nabla f \cdot \nabla g = \text{R.H.S}$$

applied to product of point functions (vector identities)

$$(1) \nabla(fg) = f \nabla g + \nabla f g$$

$$(2) \nabla \cdot (f \vec{d}) = f(\nabla \cdot \vec{d}) + \nabla f \cdot \vec{d}$$

$$(3) \nabla \times (f \vec{d}) = f(\nabla \times \vec{d}) + \nabla f \times \vec{d}$$

$$(4) \nabla \cdot (\vec{F} \cdot \vec{d}) = (\vec{F} \cdot \nabla) \vec{d} + (\vec{d} \cdot \nabla) \vec{F} + \vec{F} \times (\nabla \times \vec{d}) + \vec{d} \times (\nabla \times \vec{F})$$

$$(5) \nabla \cdot (\vec{F} \times \vec{d}) = \vec{d} \cdot (\nabla \times \vec{F}) - \vec{F} \cdot (\nabla \times \vec{d})$$

$$(6) \nabla \times (\vec{F} \times \vec{d}) = \vec{F}(\nabla \cdot \vec{d}) - \vec{d}(\nabla \cdot \vec{F}) + (\vec{d} \cdot \nabla) \vec{F} - (\vec{F} \cdot \nabla) \vec{d}$$

Proof (1):

$$L.H.S = \nabla(fg)$$

$$= \hat{i} \frac{\partial}{\partial x}(fg) + \hat{j} \frac{\partial}{\partial y}(fg) + \hat{k} \frac{\partial}{\partial z}(fg)$$

$$= \hat{i} \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right) + \hat{j} \left[f \frac{\partial g}{\partial y} + \frac{\partial f}{\partial y} g \right] + \hat{k} \left[f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right]$$

$$= f(\nabla g) + (\nabla f)g$$

$$= R.H.S$$

Proof (2):

$$L.H.S = \nabla \cdot (f \vec{d}) = \hat{i} \frac{\partial}{\partial x}(f d_x) + \hat{j} \frac{\partial}{\partial y}(f d_y) + \hat{k} \frac{\partial}{\partial z}(f d_z)$$

$$= \hat{i} \cdot \left[f \frac{\partial d_x}{\partial x} + \frac{\partial f}{\partial x} d_x \right] + \hat{j} \cdot \left[f \frac{\partial d_y}{\partial y} + \frac{\partial f}{\partial y} d_y \right]$$

$$+ \hat{k} \left[f \frac{\partial d_z}{\partial z} + \frac{\partial f}{\partial z} d_z \right]$$

$$= f \left[\hat{i} \frac{\partial d_x}{\partial x} + \hat{j} \frac{\partial d_y}{\partial y} + \hat{k} \frac{\partial d_z}{\partial z} \right] + \frac{\partial f}{\partial x} d_x + \frac{\partial f}{\partial y} d_y + \frac{\partial f}{\partial z} d_z$$

$$= f(\nabla \cdot \vec{d}) + \nabla f \cdot \vec{d}$$

Proof (ii):

$$\text{div}(\vec{F} \cdot \vec{G}) = \vec{i} \cdot \frac{\partial}{\partial x}(\vec{F} \cdot \vec{G}) + \vec{j} \cdot \frac{\partial}{\partial y}(\vec{F} \cdot \vec{G}) + \vec{k} \cdot \frac{\partial}{\partial z}(\vec{F} \cdot \vec{G})$$

$$= \vec{i} \left(\vec{F} \cdot \frac{\partial \vec{G}}{\partial x} + \frac{\partial \vec{F}}{\partial x} \cdot \vec{G} \right) + \vec{j} \left(\vec{F} \cdot \frac{\partial \vec{G}}{\partial y} + \frac{\partial \vec{F}}{\partial y} \cdot \vec{G} \right) + \vec{k} \left(\vec{F} \cdot \frac{\partial \vec{G}}{\partial z} + \frac{\partial \vec{F}}{\partial z} \cdot \vec{G} \right)$$

N.B.T $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$ $(\vec{a} \cdot \vec{c})\vec{b} = \vec{a} \times (\vec{b} \times \vec{c}) + (\vec{a} \cdot \vec{b})\vec{c}$

[Consider, $\vec{F} \times (\vec{j} \times \frac{\partial \vec{G}}{\partial x}) = (\vec{F} \cdot \frac{\partial \vec{G}}{\partial x})\vec{j} - (\vec{F} \cdot \vec{j})\frac{\partial \vec{G}}{\partial x}$

$$\Rightarrow (\vec{F} \cdot \frac{\partial \vec{G}}{\partial x})\vec{j} = \vec{F} \times (\vec{j} \times \frac{\partial \vec{G}}{\partial x}) + (\vec{F} \cdot \vec{j})\frac{\partial \vec{G}}{\partial x}$$

Similarly $(\vec{F} \cdot \frac{\partial \vec{G}}{\partial y})\vec{j} = \vec{F} \times (\vec{j} \times \frac{\partial \vec{G}}{\partial y}) + (\vec{F} \cdot \vec{j})\frac{\partial \vec{G}}{\partial y}$

$$(\vec{F} \cdot \frac{\partial \vec{G}}{\partial z})\vec{k} = \vec{F} \times (\vec{k} \times \frac{\partial \vec{G}}{\partial z}) + (\vec{F} \cdot \vec{k})\frac{\partial \vec{G}}{\partial z}$$

[Consider, $\vec{G} \times (\vec{i} \times \frac{\partial \vec{F}}{\partial x}) = (\vec{G} \cdot \frac{\partial \vec{F}}{\partial x})\vec{i} - (\vec{G} \cdot \vec{i})\frac{\partial \vec{F}}{\partial x}$

$$\Rightarrow (\vec{G} \cdot \frac{\partial \vec{F}}{\partial x})\vec{i} = \vec{G} \times (\vec{i} \times \frac{\partial \vec{F}}{\partial x}) + (\vec{G} \cdot \vec{i})\frac{\partial \vec{F}}{\partial x}$$

Similarly $(\vec{G} \cdot \frac{\partial \vec{F}}{\partial y})\vec{j} = \vec{G} \times (\vec{j} \times \frac{\partial \vec{F}}{\partial y}) + (\vec{G} \cdot \vec{j})\frac{\partial \vec{F}}{\partial y}$

$$(\vec{G} \cdot \frac{\partial \vec{F}}{\partial z})\vec{k} = \vec{G} \times (\vec{k} \times \frac{\partial \vec{F}}{\partial z}) + (\vec{G} \cdot \vec{k})\frac{\partial \vec{F}}{\partial z}$$

$$\therefore \nabla(\vec{F} \cdot \vec{G}) = \vec{F} \times (\vec{i} \times \frac{\partial \vec{G}}{\partial x}) + (\vec{F} \cdot \vec{i})\frac{\partial \vec{G}}{\partial x}$$

$$+ \vec{F} \times (\vec{j} \times \frac{\partial \vec{G}}{\partial y}) + (\vec{F} \cdot \vec{j})\frac{\partial \vec{G}}{\partial y}$$

$$+ \vec{F} \times (\vec{k} \times \frac{\partial \vec{G}}{\partial z}) + (\vec{F} \cdot \vec{k})\frac{\partial \vec{G}}{\partial z}$$

$$+ \vec{G} \times (\vec{i} \times \frac{\partial \vec{F}}{\partial x}) + (\vec{G} \cdot \vec{i})\frac{\partial \vec{F}}{\partial x}$$

$$+ \vec{G} \times (\vec{j} \times \frac{\partial \vec{F}}{\partial y}) + (\vec{G} \cdot \vec{j})\frac{\partial \vec{F}}{\partial y}$$

$$+ \vec{G} \times (\vec{k} \times \frac{\partial \vec{F}}{\partial z}) + (\vec{G} \cdot \vec{k})\frac{\partial \vec{F}}{\partial z}$$

$$= \vec{F} \times (\nabla \times \vec{G}) + \vec{G} \times (\nabla \times \vec{F}) + (\vec{F} \cdot \nabla)\vec{G} + (\vec{G} \cdot \nabla)\vec{F}$$

= R.H.S

Proof ⑤: $\nabla \cdot (\vec{F} \times \vec{a}) = \vec{a} \cdot (\nabla \times \vec{F}) - \vec{F} \cdot (\nabla \times \vec{a})$ (P.T.)

$$\text{L.H.S} = \nabla \cdot (\vec{F} \times \vec{a})$$

$$\begin{aligned} &= \vec{i} \cdot \frac{\partial}{\partial x} (\vec{F} \times \vec{a}) + \vec{j} \cdot \frac{\partial}{\partial y} (\vec{F} \times \vec{a}) + \vec{k} \cdot \frac{\partial}{\partial z} (\vec{F} \times \vec{a}) \\ &= \vec{i} \cdot \left(\vec{F} \times \frac{\partial \vec{a}}{\partial x} + \frac{\partial \vec{F}}{\partial x} \times \vec{a} \right) + \vec{j} \cdot \left(\vec{F} \times \frac{\partial \vec{a}}{\partial y} + \frac{\partial \vec{F}}{\partial y} \times \vec{a} \right) \\ &\quad + \vec{k} \cdot \left(\vec{F} \times \frac{\partial \vec{a}}{\partial z} + \frac{\partial \vec{F}}{\partial z} \times \vec{a} \right) \end{aligned}$$

w.k.T, $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$

$$\begin{aligned} &= \left(\vec{i} \times \frac{\partial \vec{F}}{\partial x} \right) \cdot \vec{a} + \left(\vec{j} \times \frac{\partial \vec{F}}{\partial y} \right) \cdot \vec{a} + \left(\vec{k} \times \frac{\partial \vec{F}}{\partial z} \right) \cdot \vec{a} \\ &\quad - \left(\vec{i} \times \frac{\partial \vec{a}}{\partial x} \right) \cdot \vec{F} - \left(\vec{j} \times \frac{\partial \vec{a}}{\partial y} \right) \cdot \vec{F} - \left(\vec{k} \times \frac{\partial \vec{a}}{\partial z} \right) \cdot \vec{F} \end{aligned}$$

$$= \underline{(\nabla \times \vec{F}) \cdot \vec{a}} - (\nabla \times \vec{a}) \cdot \vec{F}$$

= R.H.S

⑥ $\text{L.H.S} = \underline{\nabla \times (\vec{F} \times \vec{a})} = \vec{i} \times \frac{\partial}{\partial x} (\vec{F} \times \vec{a}) + \vec{j} \times \frac{\partial}{\partial y} (\vec{F} \times \vec{a}) + \vec{k} \times \frac{\partial}{\partial z} (\vec{F} \times \vec{a})$

$$\begin{aligned} &= \vec{i} \times \left(\vec{F} \times \frac{\partial \vec{a}}{\partial x} + \frac{\partial \vec{F}}{\partial x} \times \vec{a} \right) \\ &\quad + \vec{j} \times \left(\vec{F} \times \frac{\partial \vec{a}}{\partial y} + \frac{\partial \vec{F}}{\partial y} \times \vec{a} \right) \\ &\quad + \vec{k} \times \left(\vec{F} \times \frac{\partial \vec{a}}{\partial z} + \frac{\partial \vec{F}}{\partial z} \times \vec{a} \right) \end{aligned}$$

w.k.T $\vec{a} \times (\vec{b} \times \vec{c}) \pm (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

$$\begin{aligned} &= \left(\vec{i} \cdot \frac{\partial \vec{a}}{\partial x} \right) \vec{F} - (\vec{i} \cdot \vec{F}) \frac{\partial \vec{a}}{\partial x} \\ &\quad + \left(\vec{j} \cdot \frac{\partial \vec{a}}{\partial y} \right) \vec{F} - (\vec{j} \cdot \vec{F}) \frac{\partial \vec{a}}{\partial y} \\ &\quad + \left(\vec{k} \cdot \frac{\partial \vec{a}}{\partial z} \right) \vec{F} - (\vec{k} \cdot \vec{F}) \frac{\partial \vec{a}}{\partial z} \end{aligned}$$

$$\begin{aligned}
& + \left(\vec{i} \cdot \frac{\partial \vec{F}}{\partial x} \right) \vec{a} - (\vec{i} \cdot \vec{a}) \frac{\partial F}{\partial x} \\
& + \left(\vec{j} \cdot \frac{\partial \vec{F}}{\partial y} \right) \vec{a} - (\vec{j} \cdot \vec{a}) \frac{\partial F}{\partial y} \\
& + \left(\vec{k} \cdot \frac{\partial \vec{F}}{\partial z} \right) \vec{a} - (\vec{k} \cdot \vec{a}) \frac{\partial F}{\partial z} \\
& = \vec{F}(\nabla \cdot \vec{a}) + \vec{a}(\nabla \cdot \vec{F}) - (\vec{F} \cdot \nabla) \vec{a} - (\vec{a} \cdot \nabla) \vec{F}
\end{aligned}$$

(Pb) If $\vec{a}, \vec{b}$ are irrotational then prove that $\vec{a} \times \vec{b}$ is solenoidal

Sol Suppose $\vec{a}, \vec{b}$ be irrotational

$$\Rightarrow \nabla \times \vec{a} = \vec{0} \quad \& \quad \nabla \times \vec{b} = \vec{0}$$

$$\text{Now } \nabla \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\nabla \times \vec{a}) - \vec{a} \cdot (\nabla \times \vec{b})$$

$$= \vec{b} \cdot (\vec{0}) - \vec{a} \cdot (\vec{0})$$

$$= 0 - 0$$

$$= 0$$

$\therefore \vec{a} \times \vec{b}$ is solenoidal

(Q) prove that $\nabla \cdot (\vec{r}^n \vec{r}) = (n+3)\vec{r}$

Sol L.H.S = $\nabla \cdot (\vec{r}^n \vec{r})$

$$= \vec{r} \cdot (\nabla \vec{r}^n) + \nabla \vec{r}^n \cdot \vec{r}$$

$$\text{Now } \nabla \cdot \vec{r} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 3$$

$$\nabla \vec{r}^n = \vec{i} \frac{\partial}{\partial x}(\vec{r}^n) + \vec{j} \frac{\partial}{\partial y}(\vec{r}^n) + \vec{k} \frac{\partial}{\partial z}(\vec{r}^n)$$

$$= \vec{i}(n\vec{r}^{n-1})\frac{x}{r} + \vec{j}(n\vec{r}^{n-1})\frac{y}{r} + \vec{k}(n\vec{r}^{n-1})\frac{z}{r}$$

$$\therefore \nabla \cdot (\vec{r}^n \vec{r}) = 3\vec{r}^n + (n\vec{r}^{n-2} \vec{r}) \cdot \vec{r}$$

$$= 3\vec{r}^n + n\vec{r}^n$$

$$= (3+n)\vec{r}^n$$

$$= \text{R.H.S}$$